PRE-ANALYSIS PLAN

#ManyDaughters

Impact of Having Daughters on Attitudes, Preferences, and Behaviors using SOEP data

This pre-analysis plan presents our empirical strategy for testing four hypotheses on the causal impact of having daughters on Attitudes, Preferences, and Behaviors using SOEP data. For each hypothesis, we (i) define and code all variables precisely; (ii) specify inclusion and exclusion criteria; (iii) detail the estimation methods; and (iv) provide guidelines for interpreting the estimated coefficients.

HYPOTHESIS 1

Having daughters leads to stronger support for gender equality (as measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters).

1. VARIABLES

a. DEPENDENT VARIABLE *support_gender_equality_attitudes*

The dependent variable *support_gender_equality_attitudes* is constructed using responses to the following three survey items from the 2018 wave (pl database), following the approach employed by Hamjediers (2021) and Stöcker et al. (2024).

Item name	Source	Survey question	#Valid Obs.	Unit of observation	Measurement scale
bip/bip_197_04 ; **pl/plh0298_v2**	pl	"Children below the age of 6 suffer if their mother works"	Valid: 25527	Individual	[1] Completely disagree [2-6] Scale [7] Completely agree Various forms of nonresponse or missing data: [-1] No answer [-2] Does not apply [-3] Implausible value [-4] Inadmissible multiple response [-5] Not included in this version of the questionnaire [-6] Version of questionnaire with modified filtering [-7] Only available in less restricted edition [-8] Question this year not part of survey
bip/bip_197_05 ; **pl/plh0302_v2**	pl	"A child under the age of three suffers if his or her mother works"	Valid: 25544	Individual	
bip/bip_197_06 ; **pl/plh0308_v2**	pl	"It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally"	Valid: 25570	Individual	

Specifically, the variables *pl/plh0298_v2* and *pl/plh0302_v2* are originally coded such that higher values agreement with traditional gender norms, and thus lower support for gender equality. To ensure consistent directionality across all items, these two variables are reverse-coded so that higher values indicate greater support for gender equality. The third item, *pl/plh0308_v2*, is already coded in the desired direction and therefore does not require transformation.

Following this, all three items are aligned on a 1–7 scale, where 1 represents the lowest and 7 the highest level of support for gender equality. An unweighted additive index is then constructed by summing the three items, yielding a scale that ranges from 3 to 21. To facilitate interpretability, the index is rescaled to range from 0 to 1.

b. INDEPENDENT VARIABLE *firstborn_child_female*

To estimate the causal effect of having daughters on attitudes toward gender equality, we adopt the measurement strategy employed by Greenlee et al. (2020), using cross sectional data, focusing on whether the respondent's firstborn child is female. While our theoretical interest lies in the broader question of whether having daughters influences support for gender equality, we operationalize this relationship using the sex of the firstborn child. This strategy follows a growing body of research that highlights the methodological advantages of using the sex of the first child as a quasi-experimental treatment condition[1] (Shafer & Malhotra, 2011; Sharrow et al., 2018).

Alternative measures, such as whether an individual has at least one daughter (Prokos et al., 2010) or the proportion of daughters among all children (Washington, 2008; Conley & Rauscher, 2013), are more subject to endogeneity issues due to "stopping rules" in fertility behavior (Dahl & Moretti, 2008), whereby parents may continue having children until they achieve a preferred number or sex composition. As a result, such indicators risk capturing not only the presence of daughters but also the unobserved preferences and selection processes that shape family size and structure.[2]

In contrast, the sex of the firstborn child is almost completely exogenous to parental characteristics and attitudes, thereby offering a stronger basis for causal inference. Moreover, focusing on the firstborn daughter avoids problems of multicollinearity inherent in other operationalizations, especially in contexts with small family sizes. Thus, while our broader theoretical interest concerns the impact of daughters on gender attitudes, we rely on the firstborn's sex as a methodologically sound proxy for estimating this effect.

To build the key independent variable *firstborn_child_female*, we use data from the biobirth database, which provides detailed retrospective information on the biological children of respondents. Specifically, we use the child-level variables indicating the sex

[1] Without sex-selective abortion, firstborn gender is random, so any parent-child-sex differences can be seen as causal.

[2] As in Greenlee et al. (2020), some robustness checks could explore the sensitivity of the results to alternative measures, such as using variables *at least one daughter* or *the proportion of daughters among all children*. However, since the PAP requires specifying one regression table per hypothesis, these analyses fall outside the primary scope and are mentioned here for illustrative purposes only.

(kidsex[n]) and year of birth (kidgeb[n]) of each child. These variables are linked to parental records via individual (pid) and household (cid) identifiers.

We identify the first biological child (kidsex[01]) for each respondent, and code *firstborn_child_female* as a binary indicator equal to 1 if the firstborn is female (biobirth/kidsex[01]=2, N=31915), and 0 if male (biobirth/kidsex[01]=1, N=34401).

Item name	Source	#Valid Obs.	Unit of observation	Measurement scale
biobirth/kidsex[01]	biobirth	66316	Individual	Generated Data
biobirth/kidgeb[01]	biobirth	68912		

To further examine whether the effect of having a daughter on gender attitudes varies by the child's age, we construct categorical indicators based on the age of the firstborn child in 2018, following the approach of Borell-Porta et al. (2019). Specifically, we calculate *firstborn_age_2018* by subtracting the birth year of the respondent's first child (kidgeb[01]) from the survey year (2018). We then classify the firstborn into three mutually exclusive age groups in the variable *firstborn_age_group_2018*:

- 0-5 years, coded as 0;
- 6-10 years, coded as 1;
- 11-18 years, coded as 2.

This categorization enables us to assess whether the influence of having a daughter varies across key stages, as parents may experience differing levels of gender-related concern or reflection depending on the child's age (Lane et al., 2017).

Unlike Borell-Porta et al., who focus exclusively on daughters, we define these age groups independently of the child's sex. This allows for a clearly defined reference group, respondents with a firstborn son in the same age category, enabling gender-specific comparisons. In an additional separate specification, we then interact the binary indicator for having a firstborn daughter *firstborn_child_female* with the age group category *firstborn_age_group_2018* to test whether the daughter effect varies across child age.

c. CONTROLS VARIABLES

Given that our primary independent variable, *firstborn_child_female*, is exogenous and approximates a natural experiment, we do not require the inclusion of control variables to identify an unbiased causal effect. However, following practices in quasi-experimental research (Franklin, 1991; Angrist & Pischke, 2009; Shafer & Malhotra, 2011), we include a limited set of pre-treatment control variables to improve the precision of our estimates and reduce residual variance.

We would like to include a standard limited set of "pre-treatment" control variables to account for individual-level characteristics that may influence gender attitudes independently of the sex of the firstborn child. These controls are selected based on established findings in the literature and are intended to improve the precision of our estimates without introducing post-treatment bias. Each of these control variables is theoretically motivated and detailed below, along with information on their operationalization.

Specifically, we control for respondent's:

- Sex
- Age (and age squared)
- Education level

Control	Dataset/item and coded	Description and theoretical justification	Operationalization
Sex	ppathl/sex Sex=1 (Male) Sex=2 (Female) N (valid)= 1 358 895	Indicates the respondent's biological sex Sex is a core predictor of gender attitudes, with empirical evidence showing that women tend to express more egalitarian gender views than men prior to the transition to parenthood (Brewster & Padavic, 2000; Perales et al., 2018).	Although sex is not directly generated between 2015 and 2018 in the dataset, it is likely that the longitudinal structure of the SOEP will allow us to recover this information by merging earlier waves with the 2018 sample. We will build a binary variable, *respondent_female*, coded as 1 if the respondent is female (sex = 2) and 0 if male (sex = 1).
Age (and Age^2)	ppathl/gebjahr Value (1882 from 2022) N (valid) = 1 352 458	Age may influence views on gender roles (Lynott & McCadnless, 2000; Shafer & Malhotra, 2011). Moreover, the relationship between age and gender attitudes may be non-linear.	The continuous variable (in years) *age* is calculated as: *age_2018* = 2018 - gebjahr This yields the respondent's age at the time of the 2018 survey. To allow for non-linear effects of age on gender attitudes, we also compute a squared age term *age_square_2018.* *age_square_2018*= (*age_2018*)^2
Education level	pgen/pgisced11 Value (0 to 8) N (valid) = 370 707	Higher levels of educational attainment are positively associated with more egalitarian gender attitudes (Cunningham et al. 2005).	The *education* variable, based on the highest degree of the respondent, was constructed by aggregating the eight ISCED codes into four categories:

			0 = Primary or below (ISCED [0-1]); 1 = Secondary (ISCED [2-3]); 2 = Post-secondary (ISCED [4-5]); 3 = Tertiary degree (ISCED [6-8]).

We thus decide to adopt a parsimonious approach, deliberately excluding variables that are either post-treatment or variables that could be influenced by the sex of the firstborn child. Specifically, we deliberately omit marital status, employment status, number of children, individual and household income, religiosity, and region of residence (east/west or urban/rural). While citizenship is technically pre-treatment variable, we exclude it due to expected minimal variation in the sample, which consists predominantly of German nationals.

To ensure consistency and comparability across models, we apply the same set of control variables throughout the analysis.[3]

2. INCLUSION/EXCLUSION CRITERIA

We restrict the sample to respondents aged 18-54 who are parents of at least one child under age 18 living in the household, and for household-level analyses, to heterosexual couples.

We further exclude any cases in which the sex of the firstborn cannot determined and drop observations with missing, non-applicable, or invalid codes on any key dependent, independent, or control variable.

3. EXACT ANALYSIS

The complete model takes the form:

$$X_{ih} = \beta_0 + \beta_1 F_{ij} + \sum_{a=1}^{2} \sigma_a A_{ih}^a + \sum_{a=1}^{2} \delta_a (F_{ih} \times A_{ih}^a) + \theta \boldsymbol{C}_{ih} + \varepsilon_{ih}$$

where:

X_{ih} : *support_gender_equality_attitudes*, support for gender equality attitudes, an index for individual *i* in household *h*, rescaled between 0 and 1;

F_{ih} : *firstborn_child_female*, binary variable equal to 1 if the respondent's firstborn child is female, 0 otherwise;

[3] An exception is made for Hypothesis 3, which operates at the household level. This will be discussed in detail in the relevant section.

A_{ih}^1, A_{ih}^2: *firstborn_age_group_2018* category variable for age of the first-born child. A_{ih}^1 if the firstborn is aged 6-10, A_{ih}^2 1 if aged 11 or older; the reference group is 0-5 years.

$F_{ih} \times A_{ih}^a$: *firstborn_child_female* × *firstborn_age_group_2018*, interaction terms between the firstborn daughter indicator and child age group categories, estimating heterogeneity by child age;

$\boldsymbol{C}_{ih}$: Vector of pre-treatment control variables including respondent's sex, age, age squared and educational attainment;

ε_{ih}: individual-level error term.

We implement a stepwise estimation strategy in four stages to assess the stability of the estimated effects:

1. Baseline Model: includes only the first daughter indicator F_{ij};
2. Controls Model: adds the vector of pre-treatment controls $F_{ij} + \boldsymbol{C}_{ih}$;
3. Interaction Model: includes the first-born daughter indicator, child age group categories, and their interactions $F_{ij} + A_{ih}^a + F_{ih} \times A_{ih}^a$;
4. Full Model: Full specification with both interaction terms and controls variables $F_{ij} + A_{ih}^a + F_{ih} \times A_{ih}^a + \boldsymbol{C}_{ih}$.

If we do not have enough observations necessary to detect significant effects for the interaction terms (models 3 & 4), we will not conduct these analyses.

The results from the four model specifications will be presented in a single table. Each column will correspond to a different model specification.

All models are estimated using weighted least squares (WLS). Following the methodology proposed by Goebel et al. (2019) we apply individual-level cross-sectional weights to ensure that our estimates are representative of the German population and to correct for sampling design. Specifically, we use the weight variable *phrf* (2018 wave, *biphrf*) from the *ppathl* dataset.

To address potential intra-household correlation in responses, standard errors are clustered at the household level.

To ensure the robustness of our findings, we conduct Wald tests of the joint null hypothesis that our coefficient(s) of interest (β_1, δ_a) are zero. Specifically, in the common case of testing a single coefficient, the Wald statistic will be computed as:

$$W = \left(\frac{\widehat{\beta_1}}{standard\ error\left(\widehat{\beta_1}\right)} \right)^2$$

which under the null hypothesis that $\beta_1 = 0, W$ follows a chi-square distribution. We compare W to the relevant chi-square critical value and report the corresponding p-value.

We implement this/these test(s) subgroup by subgroup, evaluating the contribution of each variable and interaction term separately.

4. INTERPRETATION OF RESULTS

Our main interest lies in estimating the coefficient β_1 in our weighted least squares (WLS) regression models, which captures the average effect of having a firstborn daughter on gender attitudes. To assess whether our sample provides sufficient statistical power, we conducted a power analysis for both the main effect and the interaction effects.

- For the main effect (models 1 and 2), assuming a small effect size of Cohen's d = 0.15. With a significance level of α = 0.05 and desired power = 0.80, we would require approximately 700 individuals per group (i.e., 1,400 total observations) to detect a statistically significant difference between respondents with firstborn daughters and those with firstborn sons.

- For models that include interaction terms between *firstborn_child_female* and *firstborn_age_group (*models 3 & 4*)*, we are particularly interested in the coefficients δ_a, which capture heterogeneity in the daughter effect across ages. To assess the power to detect such differences, we conducted a power calculation based on ANOVA with k = 6 groups (2 child sexes × 3 age categories). Assuming a small effect size of f = 0.075,[4] the result of the analysis indicates that a total sample of approximately 2,286 respondents (381 per group) would be required to achieve 80% power at the 5% significance level (Cohen, 2013).

  These power benchmarks help ensure that our study is sufficiently equipped to test both the average and heterogeneous effects of having a daughter. If we do not have enough observations necessary to detect significant effects for the interaction terms, we will not conduct these analyses.

We interpret the hypothesis to be supported if the estimated coefficient β_1 is positive and statistically significant at the 5% level (i.e., p < 0.05, weak evidence if p < 0.10) indicating that having a firstborn daughter is associated with greater support for gender

[4] f approximately equivalent to f = $d/2$.

equality. Consistent, positive, and statistically significant estimates across model specifications will be taken as stronger evidence in support of the hypothesis.

If the inclusion of interaction terms renders the main effect β_1 non-significant, but the combined effects compared to the baseline (β_0, F_{ij}, and A_{ij}) with at least one coefficient from the interaction term δ_a yields a positive and statistically significant total effect at the 5% level (i.e., $p < 0.05$, weak evidence if $p < 0.10$) for a specific subgroup, this would also be interpreted as evidence supporting the hypothesis. Concretely, for the 6–10 age group we test $\beta 1 + \delta_{6-10} > 0$ and for the 11-18 age group we test $\beta 1 + \delta_{11-18} > 0$. Finding a significant total effect for one or more age group would support the hypothesis that having a daughter increases gender-egalitarian attitudes in an age-dependent, rather than uniformly constant, manner.

HYPOTHESIS 2

Having daughters leads to more left-leaning political preferences.

That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters).

1. VARIABLES

a. DEPENDENT VARIABLE *left_political_preferences*

The binary dependent variable *left_political_preferences* is constructed using responses to the following survey item from the 2019 wave (pl database).

Item name	Source	Survey question	#Valid Obs.	Unit of observation	Measurement scale
bjp/bjp_171 ; **pl/plh0004**	pl	In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?	Valid: 24818	Individual	[0] Completely left [-9] Scale [10] Completely right Various forms of nonresponse or missing data: [-1] No answer [-2] Does not apply [-3] Implausible value [-4] Inadmissible multiple response [-5] Not included in this version of the questionnaire [-6] Version of questionnaire with modified filtering [-7] Only available in less restricted edition [-8] Question this year not part of survey

Specifically, the variable *pl/plh0004* is originally coded such that higher values indicate a stronger tendency toward the political right (1 = completely left; 10 = completely right).

Because the hypothesis is directional and binary, we are not interested in the center as an outcome. Using a three-category specification (left, center, right) would require estimating effects for *left vs. center*, *right vs. center*, and *left vs. right*. This would necessitate combining marginal effects to test the specific hypothesis, which could

introduce additional noise and complexity. Therefore, we exclude respondents who placed themselves exactly at 5 (political center).

We categorize political attitude into two mutually exclusive groups:

- Left if *pl/plh0004* = 0, 1, 2, 3, or 4, N = 8,842 (351 + 469 + 1,938 + 3266 + 2818)
- Right if *pl/plh0004* = 6, 7, 8, 9, or 10, N = 5,069 (2351 + 1730 + 755 + 110 + 123)

The resulting *left_political_preferences* will take the value 1 if the respondent is left-leaning (pl/plh0004 = 0-4) and 0 if the respondent is right-leaning (pl/plh0004 = 6-10).

b. INDEPENDENT VARIABLE

To estimate the causal effect of having a daughter on left-leaning political preferences, we construct the same variable *first_born_child_female* as in the first hypothesis.

However, since political attitudes were measured in 2019, we adjust the analysis to account for the timing of exposure. Specifically, we create the variable *firstborn_age_2019*, which captures the age of the respondent's firstborn child in 2019. As in Hypothesis 1, we then classify the firstborn into three mutually exclusive age groups in the variable *firstborn_age_group_2019:*

- 0-5 years, coded as 0;
- 6-10 years, coded as 1;
- 11-18 years and above, coded as 2.

c. CONTROLS

In accordance with the analytical approach employed for Hypothesis 1, we include an identical set of control variables in our analysis of the 2019 wave.

Specifically, we update the values of age-related covariates, namely, *age_2018* and *age square_2018*, to reflect respondents' age in 2019, denoted by the variables *age_2019* and *age_square_2019*. Additionally, we revise the education measure to capture any changes in respondents' educational attainment between 2018 and 2019, using the variable *education*.

The following covariates (at the respondent level) will be used as controls in the regression analyses:

- Sex, represented by the binary variable *respondent_female*;
- Age, operationalized through the continuous variables *age_2019* and age_square_2019 to allow for non-linear age effects;
- Education, captured by the updated variable *education*.

2. INCLUSION/EXCLUSION CRITERIA

The inclusion and exclusion criteria defined for Hypothesis 1 are applied identically to the test of this hypothesis.

3. EXACT ANALYSIS

The complete[5] model takes the form:

$$\Pr(Y_{ih} = 1) = \Lambda(\beta_0 + \beta_1 F_{ij} + \sum_{a=1}^{2} \sigma_a A_{ih}^a + \sum_{a=1}^{2} \delta_a (F_{ih} \times A_{ih}^a) + \theta \boldsymbol{C}_{ih})$$

where:

Y_{ih} : *left_political_preferences*, a binary measure of left-leaning political preferences, binary (0 or 1) for individual *i* in household *h*;

F_{ij} : *firstborn_child_female*, binary variable equal to 1 if the respondent's firstborn child is female, 0 otherwise;

A_{ih}^1, A_{ih}^2: *firstborn_age_group_2019* category variable for age of the first-born child. A_{ih}^1 if the firstborn is aged 6-10, A_{ih}^2 1 if aged 11 or older; the reference group is 0-5 years.

$F_{ih} \times A_{ih}^a$: *firstborn_child_female* × *firstborn_age_group_2019*, interaction terms between the firstborn daughter indicator and child age group categories, estimating heterogeneity by child age;

$\boldsymbol{C}_{ih}$: Vector of pre-treatment control variables including respondent's sex, age, age squared, educational attainment.

We implement a stepwise estimation strategy in four stages to assess the stability of the estimated effects:

1. Baseline Model: includes only the first daughter indicator F_{ij};
2. Controls Model: adds the vector of pre-treatment controls $F_{ij} + \boldsymbol{C}_{ih}$;
3. Interaction Model: includes the first-born daughter indicator, child age group categories, and their interactions $F_{ij} + A_{ih}^a + F_{ih} \times A_{ih}^a$;
4. Full Model: Full specification with both interaction terms and controls variables $F_{ij} + A_{ih}^a + F_{ih} \times A_{ih}^a + \boldsymbol{C}_{ih}$.

The results from the four model specifications will be presented in a single table. Each column will correspond to a different model specification.

[5] Note that, in a logistic regression, the error term is not included explicitly in the model specification. This is because the model assumes that the unobserved component of the latent propensity follows a standard logistic distribution.

All regressions are estimated using a weighted logistic regression approach, consistent with the method employed in Hypothesis 1. Sampling weights are drawn from the 2019 wave of the SOEP dataset (variable *bjphrf* from *ppathl*). To correct for potential intra-household correlation in responses, standard errors are clustered at the household level.

To ensure the robustness of our findings, we propose conducting joint hypothesis tests based on log-likelihood comparisons. Specifically, we assess whether our key predictor *firstborn_child_female* contributes significantly to model fit relative to a more parsimonious model that excludes this variable.

We propose to compute the log-likelihoods for each model using maximum likelihood estimation. The LL statistic will be computed as:

$$LL = 2\,(LL_{model} - LL_{restricted})$$

where:

LL_{model} is the log-likelihood of the model considered (1, 2, 3, 4) and;

$LL_{restricted}$ is the log-likelihood of the same model (1, 2, 3, 4) that omits the predictor F_{ij} .

We will also implement this/these test(s) subgroup by subgroup of interaction terms *firstborn_child_female* × *firstborn_age_group_2019*, evaluating the contribution of each interaction term to model fit separately.

4. INTERPRETATION OF RESULTS

Our main interest lies in estimating the coefficient β_1 in our weighted logistic regression models, which captures the average effect of having a firstborn daughter on political preferences To assess whether our sample provides sufficient statistical power, we conducted a power analysis for both the main effect and the interaction effects.

- For the main effect (models 1 and 2), assuming a small effect size of Cohen's d = 0.15. With a significance level of α = 0.05 and desired power = 0.80, we would require approximately 700 individuals per group (i.e., 1,400 total observations) to detect a statistically significant difference between respondents with firstborn daughters and those with firstborn sons.

- For models that include interaction terms between *firstborn_child_female* and *firstborn_age_group (*models 3 & 4*)*, we are particularly interested in the coefficients δ_a, which capture heterogeneity in the daughter effect across ages.

To assess the power to detect such differences, we conducted a power calculation based on ANOVA with k = 6 groups (2 child sexes × 3 age categories). Assuming a small effect size of f = 0.075,[6] the result of the analysis indicates that a total sample of approximately 2,286 respondents (381 per group) would be required to achieve 80% power at the 5% significance level (Cohen, 2013).

These power benchmarks help ensure that our study is sufficiently equipped to test both the average and heterogeneous effects of having a daughter. If we do not have enough observations necessary to detect significant effects for the interaction terms, we will not conduct these analyses.

We interpret the hypothesis to be supported if the estimated coefficient β_1 is positive and statistically significant at the 5% level (i.e., p < 0.05, weak evidence if p < 0.10) indicating that having a firstborn daughter is associated with left-leaning preferences. Consistent, positive, and statistically significant estimates across model specifications will be taken as stronger evidence in support of the hypothesis.

If the inclusion of interaction terms renders the main effect β_1 non-significant, but the combined effects compared to the baseline (β_0, F_{ij}, and A_{ij}) with at least one coefficient from the interaction term δ_a yields a positive and statistically significant total effect at the 5% level (i.e., p < 0.05, weak evidence if p < 0.10) for a specific subgroup, this would also be interpreted as evidence supporting the hypothesis. Concretely, for the 6–10 age group we test $\beta 1 + \delta_{6-10} > 0$ (ie.$e^{(\beta 1+\delta_{6-10})} > 1)$ and for the 11-18 age group we test $\beta 1 + \delta_{11-18} > 0$ (ie. $e^{(\beta 1+\delta_{11-18})} > 1)$. Finding a significant total effect for one or more age group would support the hypothesis, meaning higher odds of left-leaning political preferences.

[6] f approximately equivalent to f = $d/2$.

HYPOTHESIS 3

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

1. VARIABLES

a. DEPENDENT VARIABLE *gender_inequality_behavior_household*

We use three time-use items from the SOEP module (1990–2022) to construct a daily-hours-based measure of intra-household gender-equality measured by behaviors.

Item name	Source	Survey question	#Valid Obs.	Unit of observation	Measurement scale
pl/pli0043_v3 [1992-2022]	pl	[…] How many hours you spend on them on an average weekday (usually Monday to Friday). Housework such as washing, cooking, cleaning - Average weekday	Valid: 685125	Individual	In hours Various forms of nonresponse or missing data: [-1] No answer [-2] Does not apply [-3] Implausible value [-4] Inadmissible multiple response [-5] Not included in this version of the questionnaire [-6] Version of questionnaire with modified filtering [-7] Only available in less restricted edition [-8] Question this year not part of survey
pl/pli0040 [1990-2021]	pl	Hours Weekday Running Errands	Valid: 693265	Individual	
pl/pli0049_v3 [1992-2022]	pl	Performing building flat / apartment / house repairs, car repairs, gardening work - Average weekday	Valid: 692635	Individual	

Specifically, we draw on the SOEP time-use module, which asks each spouse to report hours per typical weekday spent on:

- Routine housework (“washing, cooking, cleaning”; *pl/pli0043_v3*);
- Errands (“shopping, procurements, trips to government agencies”; *pl/pli0040*);
- Repairs/gardening (“repairs at the house or flat, car repairs, gardening work”; *pl/pli0049_v3*);

Following Coltrane (2000), we classify errands as gender-neutral, routine housework as female-typed, and repairs/gardening as male-typed, so that each spouse's reported hours on these three tasks form a balanced portfolio of traditional chores. We deliberately exclude any childcare item (Sullivan 2013; Voßemer & Heyne 2019; Schulz & Raab 2023), since childcare can be perceived as partly rewarding. Consistent with Procher et al. (2018), we restrict attention to Monday-Friday chores, when opportunity costs are highest and gendered reallocation of workweek tasks is most pronounced.[7] Following standard practice, we measure gender equality in behavior using the absolute-hours (Gough & Killewald, 2011; Van der Lippe et al., 2018).

We construct our dependent variable ***gender_inequality_behavior_household*** as follows.

For each spouse $s \in \{Wife,\ Husband\}$ in household h at year t, we let:

$R_{s,h,t}$ = hours per typical workday on routine housework;
$E_{s,h,t}$ = hours per typical workday on errands;
$G_{s,h,t}$ = hours per typical workday on gardening and repairs.

We then define each spouse's total daily workweek housework hours as

$$D_{s,h,t} = R_{s,h,t} + E_{s,h,t} + G_{s,h,t}$$

For each household *h* in year *t*, we define:

gender_inequality_behavior_household = $\left|D_{W,h,t} - D_{H,h,t}\right|$

where:

$D_{W,h,t}$ = total hours per weekday that the wife reports on routine housework, errands, and repairs/gardening;

$D_{H,h,t}$ = total hours per weekday that the husband reports on routine housework, errands, and repairs/gardening.

We use the absolute value because our focus is on the size of the gap, irrespective of whether the wife or husband does more.

A smaller value of ***gender_inequality_behavior_household*** indicates greater equality (i.e., spouses' behaviors are closer), whereas higher values always indicate greater inequality.

[7] Some robustness checks could explore the sensitivity of the models to alternative definitions, for example, incorporating weekend chores or adding childcare hours. However, because the PAP requires specifying a single regression table per hypothesis, these additional specifications lie beyond our primary analysis and are mentioned here for illustrative purposes only.

b. INDEPENDENT VARIABLE

To estimate the causal effect of having a daughter on household gender equality, we adopt an event-study approach following the methodology developed by Schulz and Raab (2023) and Kleven et al. (2024). This strategy leverages panel data and predicts dependent variables at different event times relative to the birth of the first child, allowing us to trace dynamic behavioral adjustments before and after the transition to parenthood of a daughter.

Specifically, we compare levels of inequality in households with firstborn daughters versus sons across several years after and before the childbirth event.

We first begin to operationalize for each household the variable *first_born_child_female_household*. Again (as in hypothesis 1), we use data from the biobirth module to identify the first biological child within each household, coded if kidsex[01]=2, ie a female, 0 otherwise. Using kidgeb[01], we then create the variable *first_birth_year_household* as a temporal reference point for the event time specification, *first_birth_year_household* = kidgeb[01].

By merging the birth information with household-year observations using individual (*pid*) and household (*hid*) information we can assign the sex and birth year of the first child to each yearly household record.

We then define the event time variable as the difference between the year of observation and the year of first birth:

event_time ; $event_time_{ht} = year_t - first_birth_year_household_h$

The transition process is captured with a set of dummy variables as in Schulz and Raab (2023):

- *event_time_3_before*: 3 years or more before the event;
- *event_time_2_before*: 2 years before the event;
- *event_time_1_before*: 1 year before the event;
- *event_time*: transition to parenthood;
- *event_time_1_after*: 1 year after the event;
- *event_time_2_after*: 2 years after the event;
- *event_time_3_after*: 3 or more years after the event.

c. CONTROLS

We rely on the same set of theoretically motivated control variables used in previous analyses. However, since our empirical strategy for testing Hypothesis 3 will use a household fixed effects model, the structure of controls is adapted accordingly.[8] As a result, only time-varying covariates are retained in the model, while individual-level time-invariant characteristics, such as educational level, are absorbed by the household fixed effects and are therefore not separately included.

[8] Because the dependent variable is measured at the household level, we estimate household fixed effects (and not individuals) to control for all unobserved, time-invariant household characteristics.

Specifically, we control for the *age* and *age_squared* of both the mother and the father at each time point, as these vary over time and may have an influence on gender equality in the household.

2. INCLUSION/EXCLUSION CRITERIA

We apply the same inclusion and exclusion criteria as above for H1 and H2, but for H3's panel models we require each household to appear in at least three survey waves (year of the event serving as the reference, before and after).

We follow each household from up to three years before first birth through three years after first birth, covering 2003–2017.

3. EXACT ANALYSIS

To estimate the causal effect of having a firstborn daughter on gender equality in household behaviors, we adopt a household fixed effects panel regression model using an event-study approach. The model traces dynamic before and after the birth of the first child, and allows for heterogeneity by the sex of the firstborn child.

The complete model takes the form:

$$Z_{ht} = \alpha_h + \sum_{a \neq 0} \gamma_a ET_{ht}^a + \sum_{a \neq 0} \varphi_a (ET_{ht}^a \times F_h) + \theta \boldsymbol{C}_{ht} + \varepsilon_{ht}$$

where:

Z_{ht} : *gender_inequality_behavior_household*, the absolute difference in total weekday housework hours between spouses in household h in year t. Lower values indicate more equality;

α_h : the household fixed effect, controlling for all unobserved, time-invariant household characteristics;

ET_{ht}^a: a set of event-time dummies indicating time a years before of after the birth of the first child. We omit $a = 0$ (year of the childbirth), which serves as the reference category;

F_h : *firstborn_child_female*, binary variable equal to 1 if the household firstborn child is female, 0 otherwise (time-invariant);

$ET_{ht}^a \times F_h$: are interaction terms that estimate how the impact of having a daughter differs across event time periods, relative to having a son;

$\boldsymbol{C}_{ih}$: Vector of pre-treatment control variables including respondent's age and age squared;

ε_{ht} : the household error term.

We estimate a single weighted ordinary least squares (OLS) fixed effects model to account for unobserved time-invariant heterogeneity. The model is weighted using the weights provided by SOEP (variable *phrf* from *ppathl* dataset). Standard errors are clustered at the household level.

We further validate our results by performing joint Wald tests using the same Wald t-statistic approach described in Hypothesis 1 to assess the contribution of the key predictor interaction term $ET_{ht}^{a} \times F_h$ to model fit.

We will implement this/these test(s) event by event, evaluating the contribution of each interaction term separately. This allows us to identify at which event times (e.g., before or after childbirth) the gender of the firstborn child significantly affects intra-household gender inequality.

4. INTERPRETATION OF RESULTS

Our main interest lies in estimating the coefficients φ_a in our weighted least squares regression model, which captures how the effect of having a daughter (relative to a son) on household gender inequality differs across time, compared to the baseline year of childbirth.

To assess whether our sample provides sufficient statistical power, we conducted a power analysis for the main effect (ie. the interaction effect).

- To assess the power to detect such differences, we conducted a power calculation based on ANOVA with k = 14 groups (2 child sexes × 7 event dummies). Assuming a small effect size of f = 0.075,[9] the result of the analysis indicates that a total sample of approximately 3 184 household-year observations (227 per group) would be required to achieve 80% power at the 5% significance level (Cohen, 2013).

  In the event that the available sample size does not meet the statistical power requirements to detect interaction effects across all 14 subgroups, we propose an alternative specification that reduces the number of event-time dummies by dropping adjacent years. Nevertheless, each household must appear in at least three waves (spanning up to three years before first birth through three years after) in order to be included in the fixed-effects analysis.[10] This ensures sufficient within-unit variation to identify time-use changes around the transition to parenthood. Any modifications to the event-

[9] f approximately equivalent to f = $d/2$.

[10] 6 groups (2 child sexes × 3 event dummies), f = 0.075, significance level 5%, power 80% = 2286 household-year observations.

time specification will be transparently documented and justified in the final analysis.

We interpret the hypothesis to be supported if at least one of the estimated post-birth interaction coefficients φ_a is negative and statistically significant at the 5% level (i.e., $p < 0.05$, weak evidence if $p < 0.10$). Concretely, for each post-birth event, $a \in \{1,2,3\}$, we test $\varphi_a < 0$. A negative and significant φ_a suggests that, at event time a (a years after childbirth), households with a firstborn daughter exhibit lower levels of intra-household gender inequality, as measured by behavioral time-use patterns, compared to those with a firstborn son.

HYPOTHESIS 4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

This hypothesis builds directly on Hypothesis 1. Accordingly, we use the same dependent variable *support_gender_equality_attitudes*, and apply the same variable construction procedures, sample restrictions, and controls.

The aim here is to examine whether the effect of having a firstborn daughter varies depending on the sex of the respondent, i.e., whether men and women respond differently to having a daughter in terms of their gender attitudes.

1. VARIABLES

a. DEPENDENT VARIABLE

We use the same dependent variable, *support_gender_equality_attitudes* as defined under Hypothesis 1 and interact it with *respondent_female* to test whether the daughter effect differs by respondent gender.

b. INDEPENDENT VARIABLE

We use the same independent variable, *firstborn_child_female* as defined under Hypothesis 1 and interact it with *respondent_male* variable to test whether the daughter effect differs by respondent gender.

c. CONTROLS

We use the same set of controls from Hypothesis 1, namely age, age² and education.[11]

[11] Sex is no longer a control variable but serves to directly test the hypothesis.

2. INCLUSION/EXCLUSION CRITERIA

The inclusion and exclusion criteria defined for Hypothesis 1 are applied identically to the test of this hypothesis.

3. EXACT ANALYSIS

To test this hypothesis, we estimate a model that includes an interaction term between the binary indicator for firstborn daughter and a binary variable for female respondent. The model allows us to assess whether the daughter effect is moderated by the respondent's sex.

The complete model takes the form:

$$X_{ih} = \beta_0 + \beta_1 F_{ih} + \beta_2 M_{ih} + \sum_{a=1}^{2} \sigma_a A_{ih}^a + \sum_{a=1}^{2} \delta_a (F_{ih} \times A_{ih}^a) + \beta_3 (F_{ih} \times M_{ih}) + \sum_{a=1}^{2} \gamma_a (M_{ih} \times A_{ih}^a) + \sum_{a=1}^{2} \omega_a (F_{ih} \times A_{ih}^a \times M_{ih}) + \theta \boldsymbol{C}_{ih} + \varepsilon_{ih}$$

where:

X_{ih} : *support_gender_equality_attitudes*, support for gender equality attitudes, an index for individual *i* in household *h*, rescaled between 0 and 1;

F_{ih} : *firstborn_child_female*, binary variable equal to 1 if the respondent's firstborn child is female, 0 otherwise;

M_{ih} : *respondent_male*, binary variable equal to 1 if the respondent's is a male, 0 otherwise;

A_{ih}^1, A_{ih}^2: *firstborn_age_group_2018* category variable for age of the first-born child. A_{ih}^1 if the firstborn is aged 6–10, A_{ih}^2 1 if aged 11 or older; the reference group is 0–5 year;

$F_{ih} \times A_{ih}^a$: *firstborn_child_female* × *firstborn_age_group_2018*, interaction terms between the firstborn daughter indicator and child age group categories, estimating heterogeneity by child age;

$F_{ih} \times M_{ih}$: *firstborn_child_female* × *respondent_male,* Interaction between firstborn daughter and respondent sex, ie. our main coefficient of interest.

$M_{ih} \times A_{ih}^a$: *respondent_male* × *firstborn_age_group_2018,* Interaction between respondent sex and age group.

$F_{ih} \times A_{ih}^a \times M_{ih}$: *firstborn_child_female* × *firstborn_age_group_2018* × *respondent_male,* a three-way interaction that allows us to test whether the heterogeneity by daughter age varies by respondent sex, i.e., whether fathers and

mothers react differently to having daughters of different ages subgroups, our second group of coefficients of interest.

$\boldsymbol{C}_{ih}$: Vector of pre-treatment control variables including respondent's age, age squared and education;

ε_{ih}: individual-level error term.

We implement a stepwise estimation strategy in four stages to assess the stability of the estimated effects:

1. Model 1: includes only the interaction term $F_{ih} \times M_{ih}$ (without A_{ih}^{a})
2. Controls Model: adds the vector of pre-treatment controls $\boldsymbol{C}_{ih}$;
3. Model 3: includes the third term interaction $F_{ih} \times A_{ih}^{a} \times M_{ih}$;
4. Full Model: Full specification with both interaction terms and controls variables.

The results from the four model specifications will be presented in a single table. Each column will correspond to a different model specification.

If our sample lacks sufficient observations to reliably estimate the three-way interaction terms in Models 3 and 4, we will omit those specifications from our analysis.

All models are estimated using weighted least squares (WLS). Following the methodology proposed by Goebel et al. (2019) we apply individual-level cross-sectional weights to ensure that our estimates are representative of the German population and to correct for sampling design. Specifically, we use the weight variable *phrf* (2018 wave, *biphrf*) from the *ppathl* dataset.

To address potential intra-household correlation in responses, standard errors are clustered at the household level.

To ensure the robustness of our findings, we conduct Wald tests of the joint null hypothesis that our coefficient(s) of interest (β_3, ω_a) are zero. Specifically, in the common case of testing a single coefficient, the Wald statistic will be computed as:

$$W = \left(\frac{\widehat{\beta_3}}{standard\ error\left(\widehat{\beta_3}\right)} \right)^2$$

which under the null hypothesis that $\beta_3 = 0, W$ follows a chi-square distribution. We compare W to the relevant chi-square critical value and report the corresponding p-value.

We implement this/these test(s) subgroup by subgroup, evaluating the contribution of each variable and interaction term separately.

4. INTERPRETATION OF RESULTS

Our main interest lies in estimating the coefficient β_3 in our weighted least squares, which captures which captures whether the daughter effect on *support_gender_equality_attitudes* differs between men and women.

To assess whether our sample provides sufficient statistical power, we conducted a power analysis for both the main effect and the interaction effects.

- For the main effect (models 1 and 2), assuming a small effect size of f = 0.075, a significance level of α = 0.05 and desired power = 0.80, we would require approximately 485 individuals per group (i.e., 1940 total observations) to detect a statistically significant difference between respondents with firstborn daughters and those with firstborn sons.

- For models that include interaction terms between *firstborn_child_female*, *respondent_male* and *firstborn_age_group (*models 3 & 4*)*, we are particularly interested in the coefficients ω_a, which capture heterogeneity in the daughter effect across ages and respondent's sex. To assess the power to detect such differences, we conducted a power calculation based on ANOVA with k = 12 groups (2 child sexes × 3 age categories x 2 respondent sexes). Assuming a small effect size of f = 0.075,[12] the result of the analysis indicates that a total sample of approximately 2988 respondents (249 per group) would be required to achieve 80% power at the 5% significance level (Cohen, 2013).

  If we do not have enough observations necessary to detect significant effects for the interaction terms *firstborn_child_male* x *respondent_female* x *firstborn_age_group*, we will not conduct these analyses.

We interpret the hypothesis to be supported if the estimated coefficient β_3 is positive ($\beta_3 > 0$) and statistically significant at the 5% level (i.e., p < 0.05, weak evidence if p < 0.10) indicating that the daughter effect is stronger among male respondents ($\beta_1 + \beta_3 > \beta_1$). Consistent and statistically significant interaction effects across specifications would be taken as stronger evidence in support of the hypothesis.

If the full model includes with additional interactions with child age (e.g., *firstborn_child_female* × *respondent_male* × *firstborn_age_group*), we interpret the hypothesis to be supported if at least one of one coefficient ω_a for a specific subgroup from these interaction terms is positive and statistically significant (i.e., p < 0.05, weak evidence if p < 0.10).Concretely, for the 6-10 age group we test $\beta_3 + \omega_{6-10} > 0$ and for the 11-18 age group we test $\beta_3 + \omega_{11-18} > 0$. Finding a significant total effect for

[12] f approximately equivalent to f = $d/2$.

one or more age group would support the hypothesis. This would suggest that the daughter effect differs by both respondent sex and child developmental stage, rather than being uniform.

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